

Batul Mankada

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EXPERIENCE

Associate Technical Consultant <i>Oracle OFSS</i>	Jul 2025 – Present <i>Bangaluru, India</i>
- Developing automation tools to analyze retrofit hotfix invasive changes across enterprise banking modules. - Building intelligent workflows to streamline impact analysis, patch validation, and SDLC processes. - Working on Java-based banking systems and contributing to AI-assisted automation initiatives within OFSS products.	
SDE (ML/AI) <i>Yactraq Online Inc.</i>	Jun 2024 – Jun 2025 <i>Remote</i>
- Built real-time emotion and gender detection systems using acoustic features with OpenSMILE and DeepGram. - Deployed Whisper.cpp on-prem and fine-tuned Whisper models using LoRA for multilingual call transcription. - Conducted A/B testing on NVIDIA NeMo and HuggingFace ASR models for English and Spanish speech pipelines.	
Junior Backend Developer <i>Immence</i>	Mar 2023 – Aug 2023 <i>Vadodara, India</i>
- Developed backend APIs using Express.js and Nest.js with Stripe payment integration. - Integrated WhatsApp Chat and Twitter DM into a unified communication platform.	

PROJECTS

Banking Knowledge Copilot Python, LangChain, FastAPI, RAG, LLMs	2025
- Developed a RAG-based banking knowledge assistant using LLMs, vector embeddings, and semantic search. - Implemented document ingestion, embedding pipelines, and context-aware response generation.	
AI News Agent Python, LangChain, Playwright, AI Agents	2025
- Built an autonomous AI news agent that aggregates, summarizes, and categorizes real-time news using LLM workflows. - Implemented multi-agent orchestration, scraping pipelines, and AI-powered summarization workflows.	
Multi-Modal Image Retrieval PyTorch, CLIP, Deep Learning	2025
- Developed a multimodal retrieval system using image-text embeddings for semantic image search. - Implemented feature extraction and similarity matching using deep learning embedding pipelines.	
Quantum LM with Entanglement Embedding Deep Learning, NLP	2023
Explored quantum-inspired contextual embeddings using entanglement-based neural representations for semantic dependency modeling.	
Analysis on Kisan Call Center Data Hadoop, PySpark, NLP	2024
Built NLP pipelines using Hadoop and PySpark to identify and reduce duplicate agricultural queries using TF-IDF and cosine similarity.	

PUBLICATION

TRANSFORM-X: Transfer via Feature Correspondence Springer LNCS, LOD 2025	2025
- Proposed an interpretable transfer learning framework using SHAP, LIME, and feature importance-based feature correspondences across text, image, and audio domains. - Achieved 3–5% improvement in target-domain accuracy over direct transfer baselines on XNLI, Office-31, and CREMA-D datasets.	

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript
AI/ML: Machine Learning, Deep Learning, NLP, LLMs, RAG, AI Agents, Transformers, Fine-tuning
Frameworks & Tools: SpringBoot, PyTorch, TensorFlow, LangChain, FastAPI, Docker, Git, Hadoop, PySpark
Databases & Cloud: Postgres, MongoDB, MySQL, AWS, Supabase
Core CS: Data Structures & Algorithms, DBMS, OS, SDLC, System Design, Low-Level Design (LLD)

EDUCATION

M.Tech in AI & DS <i>Mahindra University — CGPA: 9.21/10 (Gold Medalist)</i>	2023 – 2025 <i>Hyderabad, India</i>
Bachelor of Engineering <i>Government Engineering College — CGPA: 8.15/10</i>	2018 – 2022 <i>Bhavnagar, India</i>